
SMART INDIA HACKATHON 2026

DETAILED PROJECT REPORT

PRAMAANAI

AI-Based Fake Identity & Document Screening System

Project Information	Details
Problem Statement ID	26188
Problem Statement	AI-Based Fake Identity & Document Screening System
Theme	Blockchain and Cybersecurity
PS Category	Software
Team Name	AlphaX
Project Type	Prototype / Technical Demonstration

Submitted as a technical project report for Smart India Hackathon 2026.

Note: This report describes the prototype architecture and capabilities represented in the supplied project material. Production deployment would require authorization, validated datasets, and integration with appropriate official systems.

CERTIFICATE

This is to certify that the project report entitled “PramaanAI — AI-Based Fake Identity & Document Screening System” has been prepared by Team AlphaX for the Smart India Hackathon 2026 problem statement identified as Problem Statement ID 26188. The report presents the proposed architecture, implementation approach, security model, testing plan, feasibility considerations, limitations, and future scope of the prototype.

The report is intended for academic and hackathon evaluation purposes and does not represent authorization for operational access to government, law-enforcement, immigration, or border-control databases.

DECLARATION

We declare that the technical description in this report is intended to accurately distinguish implemented prototype components from planned or future integrations. Any reference to external government or international systems is presented as contextual research and not as a claim of direct production integration.

ACKNOWLEDGEMENT

We express our gratitude to the Smart India Hackathon ecosystem, our institution, mentors, faculty members, and all contributors whose guidance and technical resources supported the development of PramaanAi. We also acknowledge the standards and public technical resources consulted for understanding machine-readable travel documents, border verification, document fraud, and secure information handling.

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1. EXECUTIVE SUMMARY

PraamaanAi is an implemented prototype for evidence-based document and identity verification in checkpoint and field-verification environments, designed around an Android officer application, a FastAPI verification server and a React web console. The current build combines on-device document-region detection, OCR/MRZ processing, live-photo and liveness checks, server-side document/identity validation, a labelled mock registry, explainable results, offline synchronization and an audit-oriented integrity layer.

The verification design uses multiple complementary signals rather than treating OCR or a single AI score as proof of authenticity. The current prototype checks MRZ structure and check digits, MRZ-versus-printed consistency, QR/barcode data, configured digital signatures, dates and formats, stamp/checkpoint consistency, face/document matching, image-integrity indicators, and mock-registry relationships. Results identify the contributing checks and reasons so that an authorized officer can review the evidence.

The Android workflow captures a document or PDF, uses YOLO11n region detection and phone-side processing, performs text/MRZ checks, optionally runs blink/head-turn liveness and anti-spoofing, detects Online/Weak/Offline connectivity, and encrypts supported offline cases for later synchronization. The phone sends only the required regions/crops and phone-read text for verification rather than the full camera frame.

The central FastAPI backend runs OCR, MRZ/QR/signature checks, stamp/reference checks, face matching, image forensics, crossing rules and labelled mock-registry checks. The React web console provides Verification Desk, case review, suggested action, identity intelligence, analytics and audit views. **The integrity layer uses SHA-256-linked records with HMAC-SHA256 protection**; it is a local/mock tamper-evident ledger, not a deployed public blockchain.

The prototype was validated through 187 automated server tests and a 24/24 live end-to-end wiring check. Evaluation on the project's synthetic and local real-document sets produced measurable results, while the report explicitly records weak areas such as pixel-level tampering detection, handwriting recognition, limited script coverage and the need for field calibration. Production deployment would require authorized integrations, validated field datasets, security assessment and operational approval.

2. INTRODUCTION

2.1 Context

Border and checkpoint verification involves a combination of document inspection, identity verification, database checks and officer judgment. The environment can vary substantially between checkpoints: network availability may be inconsistent, hardware resources may differ, document types and languages may vary, and the volume of travellers can change rapidly.

International standards and law-enforcement systems illustrate why multiple verification layers are useful. ICAO Doc 9303 defines specifications for machine-readable travel documents and includes parts covering document form factors, biometrics, security mechanisms and related technologies. INTERPOL also operates systems supporting frontline checks of travel and identity documents; its SLTD database contains records for documents reported lost, stolen, revoked, invalid or stolen blank.

2.2 Project Vision

PramaanAI aims to provide a unified digital workflow in which an authorized officer can capture a document, obtain structured information, perform multiple verification checks, understand why a case received a particular status, and synchronize the case with centralized infrastructure. The design emphasizes privacy, traceability, explainability and operation under imperfect connectivity.

2.3 Scope

- Passport, visa, identity-document and driving-licence style workflows represented in the prototype.
- OCR and structured field extraction from captured documents.
- MRZ parsing/checking where a machine-readable zone is applicable.
- Document/identity matching against a prototype registry or authorized data source.
- Face capture and comparison where supported by the implemented pipeline.
- Tampering/alteration indicators where supported by the available processing modules.
- Risk indication using rule-based and/or model-derived signals.
- Local cache and synchronization workflows for appropriate previously verified information.
- Centralized web monitoring and audit-oriented views.
- Authentication, role separation and secure data handling.

2.4 Out of Scope

- Direct access to restricted government databases without formal authorization.
- Automatic legal determination that a person is a criminal or fraudulent.
- Guaranteed detection of every sophisticated counterfeit or manipulation.
- Replacing the final judgment of an authorized officer.
- Storing sensitive identity information on a public blockchain.

3. PROBLEM STATEMENT AND BACKGROUND

The project addresses the problem of screening potentially fraudulent identities and documents in environments where verification must be fast, repeatable and secure. The supplied problem framing identifies altered passports or visas, forged stamps, modified personal details, impersonation, multiple identities, expired or invalid documents, and delays caused by manual inspection and basic lookups.

3.1 Core Problems

Problem Area	Operational Difficulty	PramaanAi Response
Document alteration	Text may be readable even when a document has been modified.	Combine OCR, structural checks, image/tampering indicators and officer review.
Identity impersonation	A document may belong to one person while being presented by another.	Face/document matching where supported.
Multiple identities	The same individual may appear under different names or document numbers.	Cross-case and registry matching with similarity logic.
Connectivity	Remote checkpoints may have weak or intermittent networks.	Connectivity detection, local cache and encrypted sync queue.
High volume	Repeated manual checks can slow checkpoint processing.	Mobile-first workflow and reusable verified records where appropriate.
Language/document diversity	Scripts, layouts and fields differ across document types.	Document-specific parsing and multilingual processing.
Auditability	It can be difficult to reconstruct why a case received a status.	Case records, audit logs and integrity-protected verification events.

3.2 Why a Multi-Layer Approach is Needed

A single OCR output should not be treated as proof of authenticity. OCR answers a recognition problem: what characters appear in an image? Authenticity and identity verification require additional questions: do the extracted fields conform to the expected structure; are dates internally consistent; does a machine-readable zone agree with the visible fields; does the identity match the presented person; does the record exist in the appropriate registry; and are there indicators of manipulation?

The project therefore separates extraction, validation, matching and decision support. This architecture makes it possible to explain which checks contributed to a result and to preserve an officer-in-the-loop decision process.

4. EXISTING SYSTEM AND LIMITATIONS

The project research material identifies a workflow that can depend heavily on manual inspection and basic database lookup. Such workflows remain important because trained officers can interpret physical security features and contextual information. However, digital assistance can improve consistency and reduce repetitive work when designed carefully.

- Manual document inspection can be time-consuming, particularly during periods of high throughput.
- Different checkpoints may have different hardware and network conditions.
- Basic database lookup may confirm a record without evaluating image-level or structural anomalies.
- Repeated travellers can create repetitive verification work.
- Different languages and document formats increase the complexity of field extraction.
- Sensitive identity information creates privacy and security requirements.
- An automated result without explanation can encourage over-reliance on software.

4.1 Existing Reference Landscape

The research material for the project considers ICAO document standards, INTERPOL frontline databases and document-reference resources, Indian border-management material, and open-source OCR/document-analysis projects. INTERPOL's FIELDS resource, for example, provides visual decision aids highlighting document security features, while SLTD supports checks of reported travel and identity documents.

These references inform the conceptual design but do not imply that PramaanAi is directly connected to them. Any production integration would require the appropriate official interface, authorization, data-protection controls and operational agreements.

5. OBJECTIVES

1. Develop a mobile-first screening workflow suitable for authorized field officers.
2. Extract structured information from captured identity and travel documents.
3. Apply document-specific validation rather than relying on OCR alone.
4. Support identity and document matching using exact and approximate comparison where appropriate.
5. Provide explainable risk indicators and reasons for verification outcomes.
6. Support weak-network conditions through local processing, cache and queued synchronization.
7. Protect sensitive data through encryption, access control and minimal data exposure.
8. Provide centralized monitoring, case management and audit-oriented views.
9. Maintain integrity of critical verification events using tamper-evident records.
10. Preserve the final decision with the authorized officer rather than delegating a legal decision to AI.

5.1 Design Principles

Principle	Meaning in PramaanAi
Human-in-the-loop	Software provides evidence and indicators; the authorized officer makes the final operational decision.
Privacy by design	Process locally where practical and transmit only information required for verification.
Explainability	Results include reasons/evidence rather than an unexplained pass/fail label.
Resilience	The application detects connectivity and can queue supported cases when central verification is unavailable.
Modularity	Document types and verification rules can be extended without redesigning the entire system.
Auditability	Screening and administrative events can be recorded for traceability.

6. PROPOSED SOLUTION — PRAMAANAI

PraamaanAI consists of three operational layers: an officer-facing Android application, a centralized FastAPI verification backend, and a React/TypeScript web administration console. Supporting layers include Room/WorkManager offline storage and synchronization, document-processing models, authentication/RBAC, a labelled mock registry, crossing-specific rules, and a tamper-evident audit chain.

Current implementation split: YOLO11n region detection and MiniFASNet-V2 anti-spoofing run on-device through ONNX Runtime; server-side processing includes PP-OCR/RapidOCR with Devanagari support, InsightFace face processing, OpenCV/cryptography checks and NetworkX identity relationships. The web console uses React 19, TypeScript, Vite, Tailwind CSS and Recharts.

6.1 Officer Android Application

- Captures document images and, where needed, a face image.
- Performs on-device preprocessing such as image quality checks, OCR/document parsing and preliminary analysis.
- Checks supported local cached records before initiating a server request.
- Detects connectivity state and selects online or queued processing.
- Displays verification result, risk level and reason/evidence.
- Provides a synchronization queue for pending cases.

6.2 Central FastAPI Server

- Authenticates authorized users/devices.
- Receives the minimum required encoded/processed information.
- Performs database/registry matching.
- Executes document-specific rules and risk logic.
- Coordinates face/document and identity matching where implemented.
- Returns a structured result to the mobile client.
- Records appropriate audit information.

6.3 Web Dashboard

- Displays screening statistics and recent cases.
- Supports case and flagged-record review.
- Shows officer/device activity according to role permissions.
- Provides audit logs and operational reports.
- Supports centralized monitoring of the verification environment.

7. SYSTEM ARCHITECTURE

7.1 Logical Architecture

The architecture follows a staged pipeline. The mobile client is responsible for capture and as much preprocessing as can safely be performed locally. The backend performs centralized verification that depends on registry access, shared data, or server-side computation. The dashboard consumes authorized operational data through backend services.

Layer	Major Components	Primary Responsibility
Presentation	Android Kotlin + Jetpack Compose; React 19 + TypeScript + Tailwind CSS	Officer interaction and administration.
Capture	CameraX; PDF input	Document/live-photo capture.
On-device AI	YOLO11n; ML Kit; ONNX Runtime; MiniFASNet-V2	Region detection, text/face/barcode support and anti-spoofing.
Application API	Python + FastAPI + Pydantic	Authentication, validation, orchestration and verification response.
Verification	PP-OCR/RapidOCR; Devanagari OCR; MRZ/QR/signatures; InsightFace; OpenCV	Document, identity and integrity checks.
Data	PostgreSQL; SQLite for local development	Structured records and labelled mock registry.
Offline	Room + WorkManager	Encrypted local queue and background synchronization.
Security	JWT + RBAC + Argon2 + TLS + AES-GCM/Keystore	Access control, secure transport/storage and credential protection.

7.2 Data Flow

1. Officer captures a document and, if required, a face image.
2. Device performs quality and preprocessing checks.
3. Extracted fields and supported evidence are prepared for verification.
4. Local cache is checked when a reusable verified record is available.
5. Connectivity is assessed.
6. If online, only the required processed/encoded information is sent securely to the backend.
7. If offline or weak, the supported encrypted case is placed in a local synchronization queue.
8. The backend performs registry matching, rule checks, document validation and supported identity matching.
9. The result contains status, risk indicators and reasons.
10. The officer reviews the result and makes the final operational decision.
11. Pending cases synchronize when connectivity returns, subject to duplicate-prevention and conflict handling.

8. END-TO-END WORKFLOW

The following workflow is the operational core of the system.

Stage	Process	Output
1. Capture	Document and optional face capture.	Raw image(s).
2. On-device processing	OCR, MRZ/document parsing, image quality and preliminary checks.	Structured fields + evidence.
3. Local cache check	Look for an appropriate previously verified record.	Cache hit / miss.
4. Connectivity check	Determine online, weak or offline state.	Routing decision.
5A. Online	Send required processed/encoded data securely.	Verification request.
5B. Offline/weak	Encrypt and queue supported case locally.	Pending case.
6. Server verification	Registry match, exact/fuzzy match, face/document checks, rules and risk assessment.	Structured result.
7. Result	Show verified / needs review / flagged or unable-to-verify with reasons.	Officer evidence.
8. Officer decision	Human reviews evidence and takes the authorized action.	Case decision.
9. Synchronization	Pending cases sync when connectivity returns.	Centralized case record.

8.1 Result Semantics

The current result model distinguishes Verified, Review Required, Not Verified and Official Verification Required. Advisory signals such as liveness, Devanagari extraction and electronic-document checks are shown as evidence and do not independently create a legal conclusion. The officer can Clear or Send to Admin; the admin reviews the evidence and suggested action.

9. DOCUMENT VERIFICATION METHODOLOGY

A central design choice in PramaanAi is document-specific verification. Different document types have different structures, fields and machine-readable features. Therefore, the system should route a document through a suitable validation profile.

9.1 Passport / MRTD

- OCR extracts visible fields.
- MRZ is detected and parsed where present.
- MRZ check digits/checksum rules are applied where supported.
- MRZ-derived fields are compared with visible fields.
- Dates, document number, nationality and other fields are checked for internal consistency.
- Face/photo evidence can be compared with the captured person where supported.
- Registry/status matching is performed against the available authorized or prototype data source.

ICAO Doc 9303 is the relevant standards family for machine-readable travel documents. Its published parts cover common MRTD specifications, passport specifications, machine-readable visas, biometrics, security mechanisms and PKI among other areas.

9.2 Visa

- Extract visa fields and dates.
- Validate expected field structure and formatting.
- Compare machine-readable and visible information when applicable.
- Check document/visa status against the available registry.
- Use image-level indicators as supplementary evidence rather than as the sole authenticity decision.

9.3 Driving Licence / Identity Card

- Identify the document class and expected layout.
- Extract supported fields using OCR.
- Check field consistency, dates and document identifiers.
- Apply format/layout checks against the configured document profile.
- Perform database matching where an authorized or prototype registry is available.

9.4 Why Layered Validation Matters

A successful OCR extraction only establishes that characters could be recognized. It does not establish that the document was issued by the claimed authority, has not been altered, belongs to the presenter, or is currently valid. PramaanAi therefore combines multiple independent signals and leaves final judgment to the authorized officer.

The implemented AI/ML layer combines lightweight on-device models with deterministic verification and server-side document/identity processing. The project does not treat every check as a learned model; MRZ checksums, field consistency, date rules, QR decoding, signature validation and audit hashing are deterministic controls.

Implemented model details: YOLO11n is used for document-region detection on Android (approximately 10.6 MB model); MiniFASNet-V2 provides an on-device anti-spoof signal; InsightFace is used for face processing/matching; PP-OCR/RapidOCR with a Devanagari recognizer handles document text on the server. These components are combined with rule-based checks to produce explainable evidence.

The project uses the term AI as an umbrella for image/document processing and matching components. The final report should identify each technique according to what is actually implemented rather than implying that a large trained model exists where the prototype uses deterministic processing.

Component	Purpose	Evidence Produced
OCR	Extract visible text from document images.	Text and structured candidate fields.
Document parsing	Interpret fields according to document type.	Normalized key-value data.
MRZ parsing/checking	Validate machine-readable travel-document fields where applicable.	Parsed MRZ + checksum/consistency results.
Face detection	Locate face region in a captured image.	Face bounding/quality information.
Face similarity	Compare supported face representations.	Similarity/match indicator.
Exact matching	Match identifiers exactly.	Exact match/no-match.
Fuzzy matching	Handle minor variations in names or fields.	Similarity score/possible match.
Tampering indicators	Identify image/structure anomalies where supported.	Anomaly evidence.
Risk rules	Combine verification signals.	Low/Medium/High-style indicator with reasons.

10.1 Risk Assessment

Risk and decision logic: the system exposes named contributing reasons rather than a mystery score. A possible identity match, image anomaly or registry discrepancy becomes review evidence; it does not automatically establish fraud or criminality.

10.2 Avoiding Overclaiming

- Do not report an accuracy percentage unless it was measured on a defined test dataset.
- Do not describe a model as production-grade without validation.
- Do not claim that a document is definitively counterfeit based only on an image anomaly.
- Do not claim that a person is fraudulent or criminal based only on an automated score.
- Record the dataset, test conditions and threshold whenever quantitative performance is reported.

11. MULTIPLE-IDENTITY AND MATCHING LOGIC

One of the proposed capabilities is identifying possible relationships between a person, documents and prior verification records. The objective is to surface potentially relevant matches for authorized review, not to automatically establish wrongdoing.

11.1 Matching Signals

- Exact document number match.
- Normalized name match.
- Date-of-birth match.
- Nationality or issuing-authority consistency.
- Face similarity where legally and technically supported.
- Repeated document/person relationships in the registry.
- Similarity between otherwise slightly different field representations.

11.2 Exact + Fuzzy Matching

Exact matching is appropriate for fields such as document numbers when a valid normalized representation is available. Fuzzy matching can be useful for names because transliteration, spacing and spelling can vary across scripts. However, fuzzy matching can produce false associations, so it should generate a review candidate rather than automatically merge identities.

11.3 Identity Graph Concept

The dashboard can represent relationships such as person → document → case → checkpoint → verification event. A graph-oriented view can help an authorized investigator examine repeated patterns. Such a graph should be access-controlled, purpose-limited and auditable because it can contain sensitive personal information.

12. OFFLINE, WEAK-NETWORK AND FIELD OPERATIONS

12.1 Connectivity States

State	System Behavior
Online	Complete supported verification request through the central backend.
Weak network	Use local processing/cache and queue a secure request where immediate central verification cannot reliably complete.
Offline	Perform supported on-device checks, use appropriate local records, and store an encrypted pending case for later synchronization.

12.2 Important Limitation

PraamaanAI should not be described as providing full offline access to a central registry. If a verification depends on current server-side or official registry data, that verification cannot be completed while disconnected. The application should clearly communicate the distinction between locally available evidence and centrally verified evidence.

12.3 Local Cache

The cache can reduce unnecessary repeated server calls for appropriate frequent-traveller scenarios. Cached information should have a defined validity period and provenance. Sensitive data should be encrypted, access-controlled and removed according to retention policy.

12.4 Synchronization

- 1.Create a unique case identifier.
- 2.Encrypt and persist the pending case locally.
- 3.Record the creation time and device/session context required for audit.
- 5.Retry synchronization when connectivity is restored.
- 6.Use idempotency or duplicate detection to avoid creating multiple central cases for the same queued event.

13. SECURITY AND PRIVACY — SECURE BY DESIGN

Data minimization is part of the prototype workflow: only detected regions/crops and phone-read text are prepared for verification, raw OCR text is not retained, Aadhaar numbers are masked, and evidence images are attached to a case only when admin review is required. Uploads are limited to 10 MB and 40 MP, with 1–4 images per request; oversized region coverage is rejected.

Implemented security controls include 30-minute JWT access tokens, 24-hour refresh tokens, Argon2 password hashing, login rate limiting, server-enforced RBAC, automatic session locking, encrypted token storage, TLS in transit, AES-GCM encryption for the offline queue using an Android Keystore key, request-size limits, schema validation, SHA-256 integrity hashing and HMAC-SHA256 hash-chain signing.

Security is a core requirement because identity and travel-document data is highly sensitive. The prototype therefore adopts multiple layers rather than relying on a single encryption mechanism.

Security Control	Purpose
Authentication	Restrict system access to authorized users/devices.
JWT/session security	Carry authenticated identity/permissions for API requests.
RBAC(Future work)	Separate officer and administrative capabilities.
TLS/secure transport	Protect information during transmission.
Encrypted local storage	Reduce exposure of queued/cache data.
Minimal data exposure	Transmit only information required for a verification operation.
Audit logs	Record screening and administrative activity.
Hash-based integrity	Detect changes to selected audit/verification records.
Device/session controls	Support revocation and controlled access.

BLOCKCHAIN / TAMPER-EVIDENT LEDGER

The current implementation should be described precisely as a local/mock tamper-evident ledger. Each selected verification/audit event is canonicalized, hashed with SHA-256, linked to the previous record and protected with HMAC-SHA256. The chain can be re-verified to detect alteration. No passport images, face images, biometric templates or full identity records are placed on a public blockchain.

Prototype boundary: a real distributed or permissioned blockchain network has not been deployed in the supplied build. A permissioned ledger can be evaluated in future if multiple authorities require shared integrity, subject to governance, key management, availability and privacy requirements.

The project theme includes blockchain and cybersecurity. The prototype architecture uses the blockchain concept primarily for integrity and auditability rather than storing identity data on-chain.

14.1 Proposed Integrity Mechanism

1. Create a structured verification/audit event.
2. Serialize the selected event fields in a canonical representation.
3. Compute a cryptographic hash.
4. Link the current record to the previous integrity record where a chain is used.
5. Store the resulting integrity metadata separately from sensitive identity content.
6. Recalculate and compare hashes during audit to detect unauthorized modification.

14.2 What the Ledger Protects

- Verification event integrity.
- Audit-event ordering or linkage.
- Detection of unauthorized alteration to recorded event metadata.
- Traceability of important system actions.

14.3 What Should Not Be Stored on a Public Blockchain

- Raw passport images.
- Face images or biometric templates.
- Full personal identity records.
- Sensitive travel histories.

The released prototype also uses CameraX, ML Kit, ONNX Runtime, PdfRenderer, Retrofit, WorkManager and Android Keystore. The APK was reduced from approximately 227 MB during development to about 97 MB through ARM-only native libraries and R8 shrinking.

- Unnecessary personally identifiable information.

14.4 Prototype vs Production

The current prototype should be described as a local or mock tamper-evident ledger unless a real distributed blockchain network has actually been deployed and tested. A production version could evaluate a permissioned ledger if a participating authority requires distributed integrity and multi-organization trust, but that would introduce additional governance, key-management, availability and privacy requirements.

15. ANDROID MOBILE APPLICATION

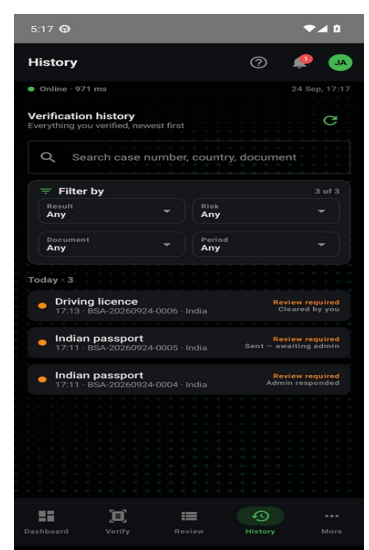
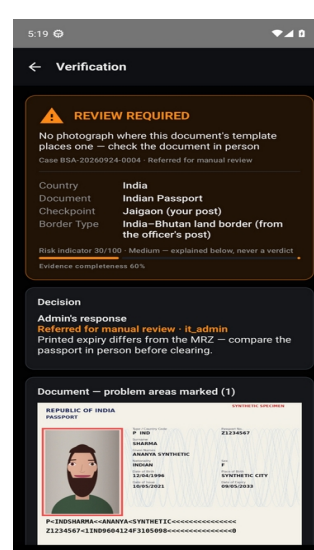
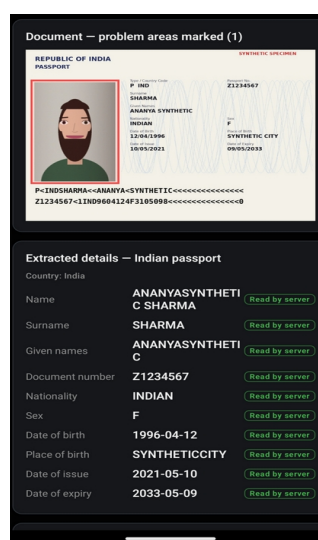
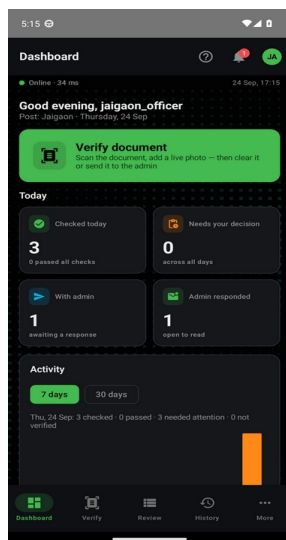
The Android application is designed as the primary field interface. The project material specifies Kotlin and Jetpack Compose, with CameraX and Room/WorkManager supporting capture and offline operations.

15.1 Main Screens / Functional Areas

Screen	Purpose
Home	Quick access to supported verification functions.
Capture Document	Capture passport/ID/visa/document image.
On-device Processing	Show OCR, parsing, face processing and preliminary checks.
Local Cache	Display appropriate cached verification records.
Verification Result	Show status, risk indicator and reasons/evidence.
Sync Queue	Show pending offline/weak-network cases and synchronization state.
Settings / Access	Provide authorized configuration and account/session functions.

15.2 Usability Principles

- Large, clear status indicators.
- Minimal number of steps during checkpoint operation.
- Immediate connectivity feedback.
- Visible reason/evidence for non-clear results.
- Clear separation between cached information and current server verification.
- Language/interface support appropriate to the deployment.



16. WEB ADMINISTRATION DASHBOARD

The web dashboard is designed for centralized monitoring and case management. The project material specifies React, TypeScript and Tailwind CSS.

16.1 Dashboard Functions

- Overview of total screening activity.
- Verified, needs-review and flagged case counts.
- Recent screening records.
- Risk distribution visualization.
- Flagged-case review.
- Reports.
- Audit logs.
- Officer activity/history.
- System settings and role-based administration.

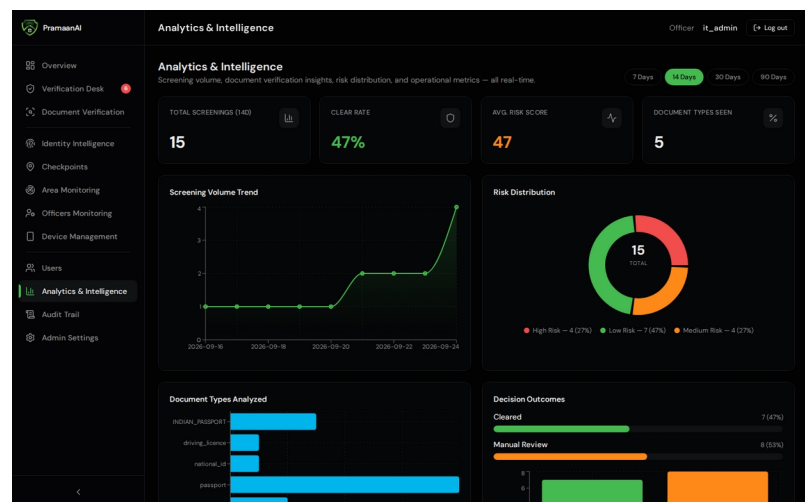
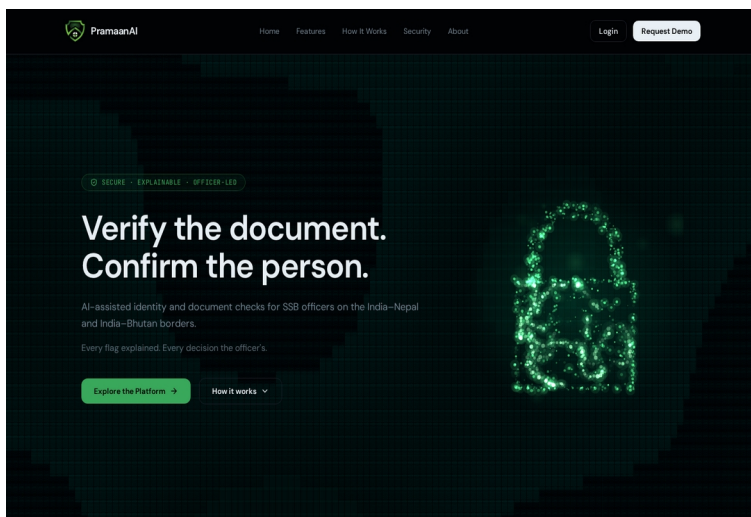
16.2 Case Management

A case record should provide a unique case identifier, timestamp, checkpoint/device context, extracted fields relevant to the case, verification signals, result status, reasons and subsequent officer action. Access to these records should follow role permissions and organizational policy.

The deployed prototype uses PostgreSQL for the central database and SQLite for local development/fallback. The live verification path was exercised against PostgreSQL 16 and fresh-database migrations were verified after fixing write-order and ID-type issues.

16.3 Monitoring

Operational monitoring can help identify unusual failure rates, synchronization backlogs, repeated device errors, and spikes in flagged cases. Monitoring should be used for system reliability and case management rather than as a substitute for human investigation.



17. BACKEND, DATABASE AND API LAYER

17.1 Backend

The backend is implemented around Python and FastAPI. FastAPI provides REST-style endpoints for authentication, document-processing requests, verification, cases, synchronization and dashboard data. The backend acts as the coordination layer between clients, processing components and the database.

17.2 Database

PostgreSQL is the primary relational database in the stated stack, with SQLite used as a local/fallback option in the prototype context. A relational schema can separate users, devices, documents, persons, cases, verification events, audit events and synchronization records.

Entity	Example Purpose
User	Officer/admin identity and role.
Device	Authorized mobile device/session information.
Person	Prototype identity record where permitted.
Document	Document metadata and identifiers.
Case	A screening/verification transaction.
Verification Event	Individual checks and results.
Audit Event	Administrative and security-relevant activity.
Sync Queue	Pending offline cases and retry status.

Backend delivered: FastAPI services, authentication, migrations, labelled mock registries, explainable decision engine and audit hash chain; verified through API documentation and automated tests.

Android delivered: on-device region detection and text/MRZ processing, live connectivity state, encrypted offline queue with synchronization, document verification flow, liveness and PDF input.

Web console delivered: Verification Desk, case review, automatic suggested action, identity intelligence, analytics, audit trail and landing page.

End-to-end integration delivered: phone → server → admin → phone workflow exercised on the live deployment.

Layer	Technology / Tool	Role
Backend	Python + FastAPI + Pydantic	REST API, validation and verification orchestration.
Database	PostgreSQL 16; SQLite	Central persistence and local development/fallback.
Android	Kotlin + Jetpack Compose + Material 3	Officer mobile application.
Camera / Input	CameraX + PdfRenderer	Document and live-photo capture; PDF input.
Offline	Room + WorkManager	Encrypted local queue and background synchronization.
On-device AI	YOLO11n + ONNX Runtime + MiniFASNet-V2	Region detection and anti-spoofing.

Text / Document	ML Kit + PP-OCR/RapidOCR + Devanagari recognizer	Text extraction and document parsing.
Identity	InsightFace + NetworkX	Face matching and identity relationships.
Web	React 19 + TypeScript + Vite + Tailwind CSS + Recharts	Admin dashboard and analytics.
Security	JWT + RBAC + Argon2 + TLS + AES-GCM + Android Keystore	Authentication, authorization and secure storage/transport.
Integrity	SHA-256 + HMAC-SHA256	Tamper-evident audit/verification chain.
Computer Vision	OpenCV + cryptography	Image analysis and signature/integrity checks.

Verification modules delivered: OCR, MRZ/QR/signature checks, stamp/reference logic, face matching, image forensics, registry checks and border crossing rules.

Offline support delivered: Online/Weak/Offline state detection, encrypted queue, background synchronization and idempotent retry behavior.

Security hardening delivered: server-enforced RBAC, TLS, Argon2, encrypted local storage, request validation, rate limiting and audit events.

1.Integrity layer delivered: SHA-256 hash-linked verification/audit records with HMAC-SHA256 protection; described as a local/mock tamper-evident ledger.

2.Deployment delivered: Dockerized services and live Render deployment used for end-to-end verification testing.

3.Testing delivered: 187 automated server tests, 24/24 live end-to-end wiring checks, Android/Python anti-spoof parity testing, and SQLite/PostgreSQL migration verification.

4.Integration: connect mobile client, backend and database through documented API contracts.

5.Verification modules: integrate OCR, document parsing, matching and configured rules.

6.Offline support: implement local cache, encrypted queue and synchronization logic.

7.Security hardening: enforce RBAC, secure transport, credential protection and audit events.

8.Integrity layer: add hash-based/tamper-evident protection for selected audit/verification events.

9.Deployment: containerize and deploy the backend/dashboard where applicable.

10.Testing: run unit, integration, security, offline and performance tests.

19.1 Development Deliverables

- Source code repository with documented setup.
- Database schema and seed/mock registry.
- Android application build.
- Web dashboard build.
- Testing was performed using the project's synthetic dataset and a local set of 57 labelled real-document photographs. Synthetic data was used for controlled workflow and check-level evaluation; the report explicitly notes that synthetic results are optimistic and should not be treated as field accuracy.
- Architecture and workflow diagrams.
- Automated server tests: 187 pass; live end-to-end wiring check: 24/24.
- Anti-spoof parity testing showed Android scores matching the Python reference within 0.05.
- Tampering detection on synthetic data achieved recall 0.33 and remains the weakest verification module.

- Server memory improved from about 685 MB to about 420 MB peak for the tested parallel verification profile.
- Six-image processing profile improved from 67.8 s to 45.6 s with identical measured accuracy in that test.
- Devanagari recognition on Nepal ID/citizenship/permit/Aadhaar samples: previously 0 characters; current confidence 0.76–0.93, with approximately +0.07 s/document.
- Phone path versus server path on 39 real captures: status 38/39; document type 39/39.
- Document-type recognition on 57 real photographs: 0.877.
- Synthetic OCR key fields: 18/18.
- Needs-attention detection on synthetic data: F1 0.925 (precision 0.939; recall 0.912).
- Expected check-level outcomes on synthetic cases: 53/56.
- Overall status matched expectation on 46 synthetic cases: 0.891.

20.1 MEASURED RESULTS

- Test cases and test evidence.
- Deployment configuration.
- Known limitations and future-work documentation.

20. TESTING AND EVALUATION PLAN

Quantitative claims should be based on reproducible tests. The following framework can be used to populate the final report with actual measurements.

Test ID	Scenario	Expected Result
T01	Clear passport image	OCR extracts expected fields.
T02	MRZ with valid check digits	MRZ validation passes.
T03	MRZ with altered check digit	Checksum inconsistency is reported.
T04	Visible field vs MRZ mismatch	Inconsistency is reported for review.
T05	Expired document in registry	Expiry/status discrepancy is shown.
T06	Known prototype flagged record	Registry match is surfaced.
T07	Different name formatting	Fuzzy matching creates a candidate, not an automatic identity merge.
T08	Face mismatch	Face comparison contributes a review/flag signal where implemented.
T09	Blurred image	Low-quality capture is detected or extraction confidence falls.
T10	Weak network	System routes to supported queue/cache behavior.
T11	Offline	Supported case is stored securely and marked pending.
T12	Reconnect	Queued case synchronizes without duplication.
T13	Unauthorized API call	Request is rejected.
T14	Officer/admin permission test	Role-restricted action is denied.
T15	Audit integrity test	Unauthorized record modification is detectable.

20.2 PERFORMANCE AND VALIDATION METRICS

- OCR processing time per image.
- End-to-end online verification time.
- Offline queue write time.
- Synchronization time after reconnection.
- API response time under representative load.
- Database query latency.
- Memory/CPU use on the Android device.
- False-positive and false-negative rates on a defined test dataset.

20.3 INTERPRETATION AND TEST LIMITATIONS

The measured values above are prototype evidence, not production accuracy claims. Synthetic documents can overestimate performance. Real-document photographs were kept locally and were not published. Tampering, handwriting and anti-spoofing require larger, representative field datasets and calibration before any such signal should become a blocking check.

21. FEASIBILITY ANALYSIS

21.1 Technical Feasibility

- The proposed stack uses established web, Android, database and API technologies.
- Mobile preprocessing can reduce network payload and latency.
- A centralized backend can serve multiple checkpoints without requiring a full server installation at each checkpoint.
- Room and WorkManager support local persistence and background synchronization patterns on Android.
- Modular verification services permit document-specific rules.

21.2 Operational Feasibility

The mobile-first design is intended to fit environments where officers already operate mobile devices. The workflow avoids assuming continuous connectivity and provides a clear boundary between local processing and server verification.

21.3 Economic Feasibility

- Uses software components that are widely available and can be deployed on existing infrastructure.
- Centralized services reduce the need for a dedicated verification server at every checkpoint.
- Cloud/VPS deployment can scale with demand.
- Actual operational cost would depend on authorized infrastructure, support, security accreditation, devices, storage and network connectivity.

21.4 Scalability

The architecture can be scaled horizontally at the API layer, with database optimization and caching for high-volume environments. Production scaling would require load testing, secure key management, observability, backup/recovery, high availability and disaster-recovery planning.

22. CHALLENGES AND MITIGATION STRATEGIES

Measured mitigation evidence: Render memory was reduced from about 685 MB to about 420 MB peak by profiling model usage, loading MediaPipe only when required, sharing the OCR engine pool, limiting verification concurrency and tuning memory allocation. Handwriting recognition was tested using multiple approaches but was not shipped because none was sufficiently reliable.

Challenge	Mitigation in PramaanAi	Remaining Consideration
Weak/no internet	Local processing, cache and encrypted queue.	Central verification cannot occur without required connectivity.
Different checkpoint infrastructure	Mobile-first architecture and centralized backend.	Device compatibility and network quality must be validated.
Frequent travellers	Appropriate cached verification records.	Cache expiry and privacy policy are required.
Language/document diversity	Multilingual OCR and document-specific parsing.	Every supported format needs testing.
Sensitive identity data	On-device preprocessing, encryption, minimal transmission.	Retention and legal policy must be defined by deployment authority.
Multiple identities	Exact/fuzzy matching and face/document signals.	Potential matches require human review.
AI over-trust	Explainable results and officer-in-the-loop decision.	Training and operational procedures remain important.
Real registry access	Prototype/mock registry for demonstration.	Production integration requires authorization.
Sophisticated forgeries	Layered document/image checks.	No automated method guarantees detection of every counterfeit.

23. IMPACT AND BENEFITS

23.1 Border / Field Officers

- Faster access to structured document information.
- Clear verification reasons rather than an unexplained binary result.
- Support across different checkpoint conditions.
- Reduced repetitive processing for appropriate frequent-traveller cases.
- Offline/weak-network workflow support.
- Final decision remains with the authorized officer.

23.2 Genuine Travellers

- Potentially shorter processing for routine cases.
- More consistent verification workflows.
- Reduced unnecessary repeated checks where policy permits cached evidence.
- Support for multilingual/document-format variation.

23.3 Command / Administration

- Central monitoring.
- Case and flagged-record visibility.
- Audit-oriented information.
- Officer/device activity monitoring according to authorization.
- Pattern analysis across cases.

23.4 Social and Governance Benefits

The intended social benefit is a more consistent screening workflow while reducing avoidable processing delays. Governance benefits include stronger traceability, role separation and auditable verification events. These are design objectives; their actual impact would need to be measured through controlled field evaluation.

23.5 Environmental Considerations

The digital workflow can reduce unnecessary paper-based processing and can operate on existing mobile hardware. Environmental impact is not a primary security metric, so any quantitative environmental claim would require separate measurement.

24. LIMITATIONS

- The prototype does not establish direct production access to government or international law-enforcement databases.
- Prototype/mock registry data cannot represent the completeness, freshness or authority of an official registry.
- Document formats vary widely and require document-specific validation profiles.
- Image-based tampering detection may fail on sophisticated manipulation or poor-quality inputs.
- Face comparison performance depends on capture quality, lighting, pose, camera and validated model/data conditions.
- Offline functionality cannot replace real-time central verification when current registry information is required.
- Fuzzy matching can create false candidate matches and therefore requires human review.
- Blockchain/tamper-evident functionality in the prototype should not be described as a production distributed blockchain unless such a network has actually been deployed.
- Operational deployment would require privacy, cybersecurity, biometric-data, retention and governance controls appropriate to the jurisdiction and authority.
- Measured limitations: synthetic tampering recall was only 0.33; handwritten visa-stamp entries were not reliably recognized (0/7 with the current OCR and 1/7 even with hand-placed crops); Bengali, Gurmukhi and Dzongkha document text is not currently read; the anti-spoof model was evaluated on only a small set and remains advisory; pixel-level tampering can be missed.

These limitations are not defects to conceal; they define the boundary between the current prototype and a production-grade system.

The project also distinguishes official verification from prototype checks: e-Visa, Aadhaar/UIDAI, DigiLocker and other official sources cannot be claimed as live integrations unless an authorized interface is actually provided and tested.

25. FUTURE SCOPE

11. **Officer assistant chatbot** using retrieval-augmented generation over approved border rules, checkpoint lists and SOPs, with citations and an offline FAQ fallback; it explains procedures but never makes the decision.
12. **Geo-fencing** using checkpoint polygons and GNSS/NavIC location to associate a verification with the officer's post and provide advisory warnings outside designated crossings.
13. **CCTV integration through authorized edge processing** for person/vehicle detection, tracking, line-crossing counts, loitering/night-movement alerts and number-plate checks; no blanket face recognition without legal authorization.
14. **Satellite connectivity support for posts** with no terrestrial network, sending small case summaries first and synchronizing full evidence later.
15. **ePassport NFC processing** of DG1/DG2 using BAC/PACE derived from the MRZ, with chip-versus-printed and chip-versus-live-photo comparison and passive authentication when an authorized trust list is available.
16. **Authorized e-Visa, UIDAI secure QR, DigiLocker and Parivahan integrations to replace prototype 'confirm on official service' guidance.**
17. **Handwriting detection and recognition** using a dedicated YOLO class and a border-form handwriting dataset.
18. **Additional document-script support for Bengali, Gurmukhi and Dzongkha, plus completion of the Dzongkha user interface with native review.**
19. **Field calibration** of anti-spoof and forensic thresholds on representative real captures before any such check is allowed to block a workflow.
20. **Security hardening:** certificate pinning/mutual TLS, Play Integrity device attestation, MFA for administrators, KMS/HSM-backed key rotation, field-level database encryption, external audit-chain timestamp anchoring, post-scoped access, security monitoring and shared rate limiting.

26. CONCLUSION

PramaanAI presents an implemented mobile-first architecture for evidence-based identity and document screening at checkpoint environments. The current prototype combines document extraction, MRZ/QR/signature validation, identity/document matching, mock-registry checks, liveness and anti-spoofing, explainable risk indicators, secure communication, offline synchronization and tamper-evident audit records within a single officer workflow.

The implementation was exercised through automated tests and a live end-to-end phone-to-server-to-admin-to-phone flow. Measured results demonstrate useful prototype capability while also exposing clear limitations: tampering detection remains weak, handwriting recognition is not reliable, some scripts are unsupported, and field calibration is still required. These results are included explicitly so that the report does not overstate prototype accuracy.

The architecture keeps the final operational decision with the authorized officer and separates sensitive identity data from the integrity ledger. The next stage is therefore not simply adding more AI, but validating the system on representative authorized data, completing security and privacy controls, integrating official services through approved interfaces, and conducting controlled field evaluation before operational use.

The project therefore provides a foundation for further validation. The next stage should focus on measurable testing, validated datasets, security assessment, official integration requirements, privacy governance and controlled field evaluation. With those conditions addressed, the modular architecture can be extended without changing the core officer workflow.

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LFW — Labeled Faces in the Wild

<http://vis-www.cs.umass.edu/lfw/>

Reference notes

Official sources were checked for the report's contextual statements about ICAO Doc 9303, INTERPOL SLTD/frontline systems, and the Sashastra Seema Bal Rules. The project-specific implementation claims in this report are based on the supplied PramaanAI project material and should be reconciled against the final repository before submission.

END OF REPORT